B.Tech Project Report for Mid Sem Evaluation

(CSPC401) on

Face Morph Attack Detection

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Abstract

Face morphing attacks represent a critical vulnerability in Automated Border Control (self-service immigration) systems, allowing multiple individuals to exploit a single biometric passport. While spatial-domain detectors effectively identify traditional landmark-warped morphs, they exhibit catastrophic failure against generative latent morphs (e.g., StyleGAN2) and degrade severely under analog print-scan conditions. To bridge this gap, this paper introduces **UID-MAD (Unified Interpretable Dual-Stream Morph Attack Detection)**, a single-image framework that explicitly targets the mathematical bifurcation of morphing physics.

UID-MAD utilizes two parallel, decoupled pathways: a spatial stream powered by a LoRA-adapted DINOv2 Vision Transformer to capture geometric blending seams, and a dedicated frequency stream utilizing a high-pass and Haar wavelet filter bank to isolate the periodic Nyquist artifacts left by generative neural upsampling. Because these two modalities compete during standard joint training, the system deploys independent ArcFace heads and combines their linear cosine margins at inference using Z-score calibrated late fusion.

Evaluated zero-shot under strict ISO/IEC 30107-3 protocols across 15 operational slices of the FRLL-Morphs dataset, UID-MAD demonstrates exceptional generalization. The frequency stream successfully recovers generative attack detection—raising StyleGAN2 ROC-AUC from a random-chance 0.4988 to 0.9399 on clean images. Furthermore, the system exploits the orthogonal nature of print-scan degradation, which destroys spatial seams but inadvertently amplifies residual frequency cues. By successfully neutralizing both algorithmic and physical evasion tactics, the calibrated dual-stream system achieves a mean ROC-AUC of 0.992 across all conditions, establishing a robust, single-pass solution suitable for real-world e-Gate deployment.

Important Keywords:

Morph Attack Detection, S-MAD, DINOv2, LoRA, ArcFace, Laplacian residual, Haar wavelet, ISO/IEC 30107-3, APCER, BPCER, FRLL-Morphs, SMDD, StyleGAN2.

1. Introduction

Electronic identity documents rely on the unstated assumption that the enrolled photograph corresponds to exactly one physical person. Face morphing violates this by blending the faces of two subjects—typically a criminal and a clean-record accomplice—into a single composite image. Because Automated Facial Recognition Systems (AFRS) are trained to allow for intra-identity variations, both subjects can successfully pass through an e-Gate using the same passport.

The report identifies that morph generation relies on two fundamentally different algorithms, necessitating a dual-approach for detection:

- **Landmark-based pipelines(OpenCV,FaceMorpher, WebMorph, AMSL):** These warp faces onto an average mesh and alpha-blend the pixels, which leaves behind visible **spatial seams** like ghosting and double pupils.
- **Generative pipelines (StyleGAN2):** These invert faces into a GAN latent space and synthesize a completely new photograph. This process leaves no spatial seams but instead imprints **periodic upsampling energy** (Nyquist checkerboard artifacts). A detector that only analyzes RGB patch geometry is structurally blind to this second family of attacks.

The project treats three recurring failures in existing Morph Attack Detection (MAD) literature as primary design constraints:

1. **Black-box decisions:** Binary scores or multi-pass heatmaps do not meet the latency or auditing requirements of a border e-Gate.
2. **Memorization on tiny real sets:** Published 0.0% error rates on small datasets do not survive cross-dataset testing.
3. **Shortcut learning:** Existing models often learn to detect JPEG compression, camera noise, or color casts instead of actual morph geometry.

To overcome these, UID-MAD is designed as a standards-compliant Single-Image Morph Attack Detection (S-MAD) system. It utilizes self-supervised spatial features (DINOv2), an explicit frequency residual stream, ArcFace angular margins, heavy anti-shortcut augmentation, and a strict cross-dataset testing protocol evaluated using ISO/IEC 30107-3 metrics.

The work is organized into two phases: Phase 1 establishes the spatial baseline and documents its collapse on generative morphs, while Phase 2 introduces the frequency stream, LoRA adaptation, and Z-score fusion to produce the final, comprehensive detection results

2. RELATED WORK

Paper:Composite_Heatmaps_Interpreting_Deep_Models_in_Morph_Attack_Detection

This research focuses on making black-box Vision Transformers (ViTs) interpretable for morph attack detection by generating composite visual heatmaps. The methodology systematically obscures specific portions of an input image—such as coarse grid tiles and critical facial landmarks like the eyes, nose, and mouth—and measures the resulting drop in the model's confidence. This occlusion technique requires no pixel-level annotations and successfully aligns with human intuition, proving that the model looks at blending seams and landmark regions to detect anomalies. The approach is entirely model-agnostic, meaning it can be applied to explain the decisions of any pre-trained deep learning network without altering its internal architecture.

Paper:Deep_Learning_Techniques_for_Detecting_Morphed_Face_Images_A_Literature_Review

This comprehensive literature review distinguishes between the operational realities of Single-Image Morph Attack Detection (S-MAD) and Differential Morph Attack Detection (D-MAD). The survey evaluates the performance of standard Convolutional Neural Networks (CNNs), Generative Adversarial Networks (GANs), and ViTs, revealing a consistent trend: while models achieve high accuracy on familiar, in-domain data, they suffer severe generalization collapses on cross-dataset benchmarks like NIST FATE MORPH and FVC-onGoing. Crucially, the survey explores the internal mechanics of identity-trained facial recognition backbones (such as FaceNet or InsightFace), documenting how their training objectives mathematically incentivize them to ignore local textures, lighting, and expressions to maximize identity matching.

Paper:Enhanced_Differential_Face_Morphing_Detection_Cross-View_Quadratic_Discriminant_Analysis_on_Hybrid_Deep-Shallow_Representation

This paper proposes a classical hybrid architecture designed to capture both high-level semantics and low-level artifacts by concatenating different feature types. The system extracts a 128-dimensional deep identity embedding using FaceNet and pairs it with a 156-dimensional shallow HASC-Beta texture feature, which captures HSV and YCbCr color representations. To classify the resulting 284-dimensional vector, the framework relies on Cross-view Quadratic Discriminant Analysis (XQDA). The study successfully demonstrates that shallow handcrafted descriptors (like LBP, BSIF, and SURF), alongside frequency transformations (DCT) and image residuals, carry highly discriminatory signals necessary for catching face morphs.

Paper:Face_Morphing_Attack_Generation_and_Detection_A_Comprehensive_Survey

This broad survey maps the complete taxonomy of both morph generation pipelines and detection strategies, synthesizing results from diverse evaluation campaigns. It specifically tracks the massive performance disparities between controlled digital environments and real-world physical channel degradations, such as the Bologna-

SOTAMD print-scan campaign where Differential-MAD EER sat at 3.36% while Single-Image EER degraded to 37.10%. The paper highlights findings from independent researchers (e.g., Dargaud, Borghi, Colbois) to establish that under honest, cross-dataset evaluation conditions, realistic Equal Error Rates (EER) for modern detection systems typically range between 0.5% and 15%

Drawbacks of previous work

- The multi-pass inference loop required to generate the heatmaps is computationally prohibitive, making it far too slow to meet the strict latency budgets of real-time Automated Border Control (ABC) e-Gates. Furthermore, because it is a post-hoc explanation tool, it never actively teaches the network the geometric shape of morphing artifacts during the training phase.
- The survey inadvertently proves that using identity-based facial recognition models (like FaceNet) is the wrong foundational prior for defense. Because these models are optimized to discard intra-identity variations, they actively assist the attack by remaining blind to the precise geometric and textural anomalies that betray a morph
- The reliance on Cross-view Quadratic Discriminant Analysis (XQDA) introduces an $O(D^3)$ computational cost, outright disqualifying the system from real-time deployment. Additionally, the architecture is highly vulnerable to physical channel degradation, failing to maintain security when digital images are physically printed and scanned.
- The survey critiques the broader literature for relying on deeply flawed evaluation protocols. It highlights that older methods frequently claim perfect 0.0% error rates or 1.0 AUC scores simply by memorizing extremely small datasets (e.g., ~38 images) without using proper cross-generator hold-outs or reporting the high-security BPCER@APCER=1% metrics required by border agencies like FRONTEX.

4. Proposed Work

The UID-MAD (Unified Interpretable Dual-Stream Morph Attack Detection) system is built on a fundamental mathematical and physical insight: face morphing attacks are not homogenous and cannot be resolved by a single feature space. Morphing pipelines bifurcate into two mathematically distinct attack classes—landmark-warped morphs (which manipulate spatial geometry and pixel coordinates via alpha blending) and generative latent morphs (which synthesize novel imagery through deep neural upsampling networks like StyleGAN2). Previous architectures failed because they attempted to process these distinct physical phenomena through a single representation stream.

The proposed framework solves this by treating the detection task as a dual-stream decomposition problem, ensuring that spatial manipulation and spectral upsampling artifacts are independently isolated, mathematically modeled, and later combined via a rigorous calibration framework.

1. How the Approach Works

The UID-MAD framework processes a single input face image $I \in \mathbb{R}^{H \times W \times 3}$ (where $H = 224, W = 224$) through two parallel, decoupled processing pathways before executing a final decision fusion.

Stream A: Spatial Vision Transformer (Targeting Landmark Morphs)

Stream A is designed to capture dense spatial patch statistics, geometric inconsistencies, and blending seams (such as double iris contours and mask perimeters).

- **Backbone:** The system utilizes a self-supervised Vision Transformer, specifically vit_small_patch14_dinov2.lvd142m (timm), containing 21.6 million parameters with an embedding dimension $D = 384$ and a patch size of 14×14. For a 224×224 input, this generates a spatial grid of $N = 16 \times 16 = 256$ patch tokens alongside a class token $\mathbf{z}_{\text{cls}} \in \mathbb{R}^{384}$.
- **Multi-Scale Patch Aggregation:** To prevent average pooling from diluting local blending seams, UID-MAD concatenates four independent 384-dimensional summaries into a comprehensive 1536-dimensional spatial vector $\mathbf{f}$:

$$\mathbf{f} = \left[\mathbf{z}_{\text{cls}} \parallel \boldsymbol{\mu}_{\text{patch}} \parallel \boldsymbol{\sigma}_{\text{patch}} \parallel \mathbf{z}_{\text{max}}\right] \in \mathbb{R}^{1536}$$

Where the components are computed across all N patches as:

$$\boldsymbol{\mu}_{\text{patch}} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{z}_i\,, \quad \boldsymbol{\sigma}_{\text{patch}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\mathbf{z}_i - \boldsymbol{\mu}_{\text{patch}}\right)^2 + \varepsilon}, \quad \backslash \boldsymbol{none}\mathbf{z}_{\text{max}} = \max_i \mathbf{z}_i$$

Explanation of Math: $\boldsymbol{\sigma}_{\text{patch}}$ serves as an explicit mathematical seam detector, because pixel-level blending discontinuities inherently elevate patch-wise feature variance. $\mathbf{z}_{\text{max}}$ preserves peak local anomalies that mean pooling would otherwise wash out.

- **Low-Rank Adaptation (LoRA):** To adapt the frozen self-supervised manifold to forensic tasks without destroying its pre-trained geometric priors, LoRA factors are injected into the attention and MLP layers of the last six ViT blocks (blocks 6–11):

$$\Delta\mathbf{h} = \frac{\alpha}{r}\,\mathrm{Dropout}(\mathbf{x})\,W_A^{\top}W_B^{\top}, \quad \text{where } r = 16,\ \alpha = 32,\ \frac{\alpha}{r} = 2$$

- **Spatial Projector:** A two-layer bottleneck network maps $\mathbf{f}$ down to the final spatial embedding $\mathbf{e}_{\text{spatial}} \in \mathbb{R}^{256}$ via GELU activations and Batch Normalization.

Stream B: Frequency Residual Extractor (Targeting Generative Morphs)

Stream B completely ignores identity and RGB geometry, focusing entirely on high-frequency spectral signatures and periodic upsampling artifacts (such as Nyquist checkerboards) left by generative models.

- **Filter Bank & Residual Extraction:** The image is passed through a non-trainable, depthwise 3×3 filter bank combining an 8-neighbor Laplacian kernel K_{Lap}, a horizontal second-difference kernel K_h, and a vertical second-difference kernel K_v:

$$K = \frac{1}{3}\left(K_{\text{Lap}} + K_h + K_v\right), \quad R = |I * K|$$

- **Haar Wavelet Sub-band:** To isolate diagonal checkerboard patterns typical of GAN upsampling, a rectified 2-D Haar high-high sub-band $HH(x, y)$ is computed and bilinearly upsampled to 224×224:

$$HH(x,y) = \frac{1}{2}\left(I_{2x,2y} - I_{2x,2y+1} - I_{2x+1,2y} + I_{2x+1,2y+1}\right)$$

The composite frequency residual map C is formed by blending both signals:

$$C = 0.5\,R + 0.5\,|HH|_{\uparrow}$$

- **Frequency ConvNet:** A 4-stage convolutional network ($3 \to 32 \to 64 \to 128 \to 128$, utilizing stride-2 convolutions, BatchNorm, and GELU activations) processes C into an adaptive pool, generating the frequency embedding $\mathbf{e}_{\text{freq}} \in \mathbb{R}^{128}$.

Decoupled Training & Z-Score Calibrated Fusion

- **Additive Angular Margin (ArcFace):** All embeddings and class weights are ℓ_2-normalized to enforce angular separation, optimizing the classification boundary using scale $s = 32.0$ and angular margin $m = 0.25$:

$$\hat{\mathbf{e}} = \frac{\mathbf{e}}{\|\mathbf{e}\|_2}, \quad \hat{\mathbf{w}}_j = \frac{\mathbf{w}_j}{\|\mathbf{w}_j\|_2}, \quad \cos\theta_j = \hat{\mathbf{w}}_j^{\top}\hat{\mathbf{e}}$$

$$\mathcal{L}_{\text{ArcFace}} = -\log\frac{e^{s\phi_y}}{e^{s\phi_y} + \sum_{j \neq y} e^{s\cos\theta_j}}, \quad \text{where } \phi = \cos(\theta_y + m)$$

- **Decoupled Multi-Task Loss:** To prevent modal competition (where fast-converging CNN streams dominate slow ViT gradients), three independent heads are optimized simultaneously during training:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ArcFace}}(\mathbf{e}_{\text{fuse}}, y) + 0.5\,\mathcal{L}_{\text{ArcFace}}(\mathbf{e}_{\text{spatial}}, y) + 0.5\,\mathcal{L}_{\text{ArcFace}}(\mathbf{e}_{\text{freq}}, y)$$

- **Z-Score Calibrated Decision Fusion:** At inference, raw softmax scores saturate and fail. Instead, the system computes the linear cosine margin for each stream (S_s and S_f), standardizes them against the mean (μ_{bf}) and standard deviation (σ_{bf}) of genuine bona fide images within the active evaluation slice, and fuses them with equal weights:

$$Z_s = \frac{S_s - \mu_{\text{bf},s}}{\sigma_{\text{bf},s} + \varepsilon}, \quad Z_f = \frac{S_f - \mu_{\text{bf},f}}{\sigma_{\text{bf},f} + \varepsilon}, \quad Z_{\text{final}} = 0.5\,Z_s + 0.5\,Z_f$$

2. Why This Approach Is Better

- **Rejection of Flawed Identity Priors:** Traditional models rely on facial recognition backbones (like FaceNet or InsightFace) which are mathematically optimized to discard local texture, lighting, and expressions to achieve identity invariance. Because those discarded dimensions are precisely where morph artifacts live, identity models inherently help the attack. UID-MAD uses self-supervised DINOv2 features, ensuring the model evaluates raw geometry and variance rather than facial identity.

- **Bifurcated Physics Handling:** Rather than forcing a single network to learn conflicting visual features, separating spatial seams into Stream A and frequency spectra into Stream B ensures that both landmark-warped composites and neural-synthesized latent morphs receive dedicated mathematical tracking.

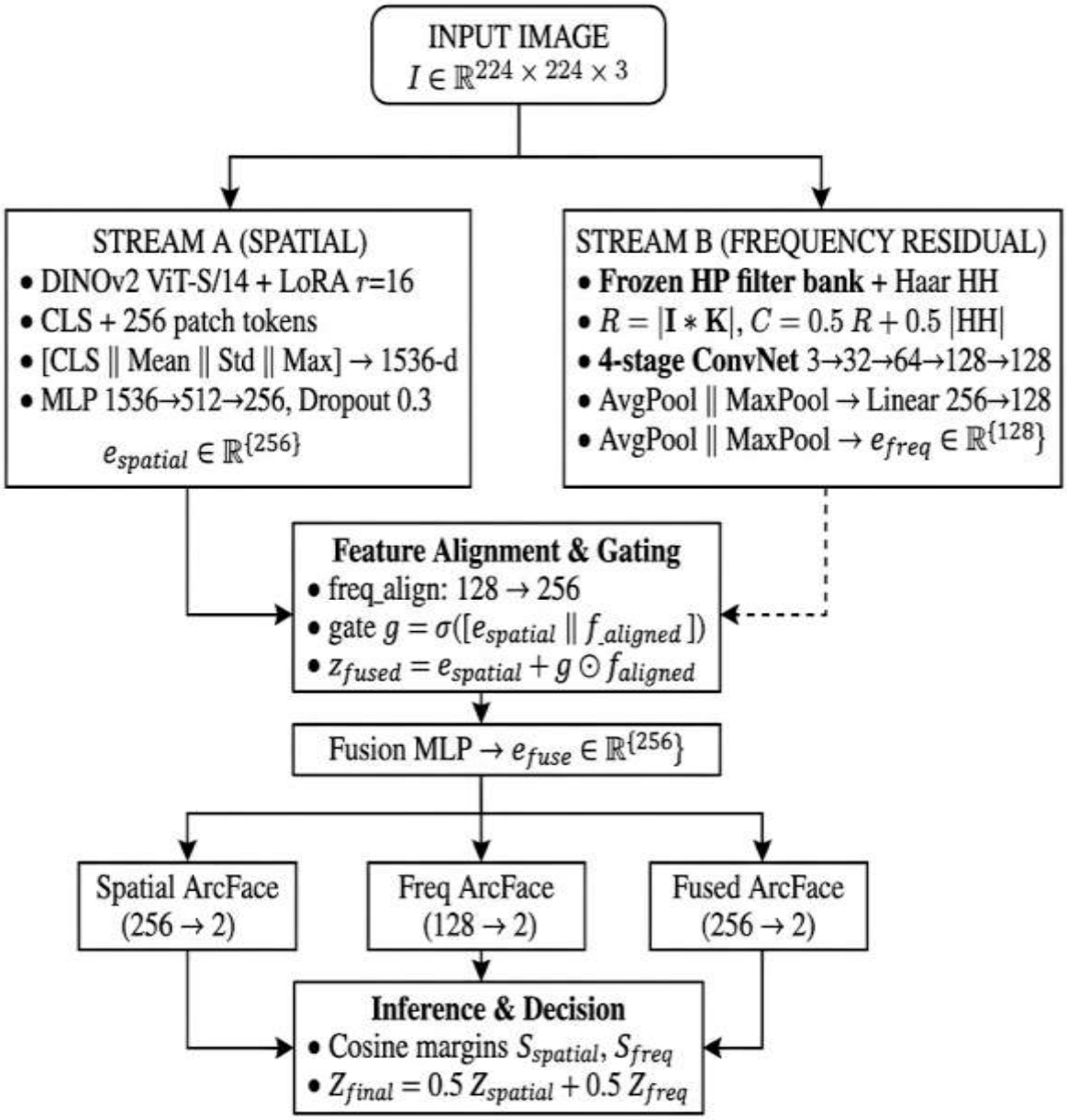
3. How It Solves Previous Problems

- **Solving the Generative Blindspot (StyleGAN2 Collapse):** In Phase 1 testing, a spatial-only ViT operated at random chance against generative StyleGAN2 morphs, scoring a catastrophic ROC-AUC of 0.4988 because generative morphs contain zero spatial seams. By integrating Stream B's high-pass filter bank and Haar wavelets, the framework captures periodic upsampling energy, lifting StyleGAN2 detection to an ROC-AUC of 0.9399 on clean images and 0.9854 under print-scan conditions.

- **Overcoming the Print-Scan Degradation Trap:** Physical print-scan processes act as analog low-pass filters that obliterate spatial seams (causing spatial-only models to fail completely, e.g., dropping WebMorph spatial AUC down to 0.6355). However, this analog modulation paradoxically enhances high-frequency noise and printer grain, boosting Stream B's frequency AUC to > 0.997 across all generators. Because their failure modes are **orthogonally complementary**, the Z-score fusion guarantees that when spatial cues fail, frequency signatures rescue the decision.

- **Eliminating Modality Competition and Score Saturation:** Early feature concatenation causes convolutional branches to overpower transformer layers, resulting in feature inversion (Trial 1 AUC dropping to 0.2422). By decoupling training heads and utilizing bona fide-calibrated Z-score fusion at inference, UID-MAD avoids probability saturation and achieves a robust mean ROC-AUC of 0.992 across all 15 ISO/IEC 30107-3 evaluation slices.

4.2 System Architecture:



4.3 Model Pipeline:

Data Ingestion and Splits

- The pipeline ingests the synthetic SMDD dataset for training and validation, utilizing a 32,000-image corpus split 80/20 into 20,030 bona fide and 11,970 morph images.
- Evaluation is strictly zero-shot, utilizing the FRLL-Morphs dataset containing 7,274 images across five distinct generators.

Training Augmentations

- Inputs are resized to 224x224 pixels and normalized using standard ImageNet statistics.
- An Albumentations pipeline applies heavy anti-shortcut transformations, including JPEG compression (quality 20–80), Gaussian and motion blur, and print-scan proxies consisting of downscaling combined with Gaussian noise.
- Further augmentations include random gamma adjustments, camera PRNU noise proxies, ToGray conversion to eliminate hue shortcuts, and CoarseDropout.

Two-Stage Optimization

- **Stage 1 (Head Warmup):** Across epochs 1 through 4, the DINOv2 backbone and LoRA weights remain frozen. The optimizer (AdamW) updates only the spatial projector, frequency ConvNet, gate, fusion MLP, and three ArcFace heads.
- **Stage 2 (LoRA Fine-Tune):** During epochs 5 through 8, the LoRA parameters on the last six ViT blocks are unfrozen and optimized alongside the heads using cosine-annealed learning rates.

Evaluation Degradations

- Slices are evaluated under three conditions: clean images, JPEG compressed images (quality 25–35), and print-scan simulated images (downscaled with JPEG compression and Gaussian noise).

Inference Path

- A locked checkpoint processes images to output spatial, frequency, and fused scores.
- The evaluation routine estimates the mean and standard deviation of spatial and frequency cosine margins exclusively on the bona fide subset of each evaluation slice.
- These statistics are used to compute an equal-weight Z-score calibrated decision ($Z_{\text{final}} = 0.5Z_s + 0.5Z_f$), which serves as the final score for ISO/IEC 30107-3 metric calculations (EER, BPCER@1%, BPCER@5%, ROC-AUC).

Engineering Constraints

- To prevent CUDA IPC deadlocks and OpenMP oversubscription on Kaggle hardware, the pipeline forces NUM_WORKERS = 0 and single-threaded execution.
- A 90-minute runtime ceiling ensures evaluation completes, with the official Phase 2 pipeline taking 58.4 minutes for training and 9.5 minutes for evaluation on dual T4 GPUs.

4.4 Proposed Algorithm:

Algorithm 1: UID-MAD Training

Input: SMDD train set D_{tr}, validation set D_{val}, config C **Output:** checkpoint θ^*

1. seed_everything(42)
2. Build MorphDetector:
    - Freeze DINOv2; inject LoRA ($r = 16$, $\alpha = 32$) on the last 6 blocks (LoRA still frozen).
    - Initialize FrequencyResidualStream with frozen K.
    - Initialize projectors, gate, fusion, and three ArcFace heads.
3. Apply WeightedRandomSampler on D_{tr}.
4. **Stage 1 (Head Warmup):** For epoch = 1 to 4
    - For each batch (I, y):
        - $(l_{fuse}, l_s, l_f) \leftarrow$ MorphDetector(I, y)
        - $L \leftarrow \text{CE}(l_{fuse}, y) + 0.5\text{CE}(l_s, y) + 0.5\text{CE}(l_f, y)$
        - AMP backward; AdamW step (updating heads + Stream B + fusion only)
    - Evaluate calibrated Z on D_{val}; save if EER improves.
5. Unfreeze LoRA parameters.
6. **Stage 2 (LoRA Fine-tune):** For epoch = 5 to 8
    - Repeat inner loop with cosine-annealed learning rates.
    - Evaluate; save best θ^* based on EER, then validation loss.
7. Return θ^*.

Algorithm 2: MorphDetector Forward (Train)

Input: mini-batch I, labels y

1. $tokens \leftarrow$ DINOv2_LoRA(I) (yields CLS, 256 patches)
2. $f \leftarrow$ [CLS ‖ mean ‖ std ‖ max] (1536-d)
3. $e_s \leftarrow$ SpatialProjector(f) (256-d)
4. $C \leftarrow 0.5|\text{HP}(I)| + 0.5\text{upsample}(|\text{Haar_HH}(I)|)$
5. $e_f \leftarrow$ FreqConvNet(C) (128-d)
6. $f_a \leftarrow$ FreqAlign(e_f) (256-d)

7. $g \leftarrow \sigma\big(\text{Gate}([e_s \parallel f_a])\big)$
8. $e_{fuse} \leftarrow \text{FusionMLP}(e_s + g \odot f_a)$ (256-d)
9. Return $\text{ArcFace}\big(e_{fuse}, y\big)$, $\text{ArcFace}(e_s, y)$, $\text{ArcFace}\big(e_f, y\big)$

Algorithm 3: Calibrated Inference and ISO Metrics
Input: locked θ^*, evaluation loader L with labels $y \in \{0,1\}$

1. For each batch, collect S_s, S_f (cosine morph − cosine bona fide).
2. On bona fide index set $B = \{i: y_i = 0\}$:
    - $\mu_s, \sigma_s \leftarrow$ mean, std of $S_s[B]$
    - $\mu_f, \sigma_f \leftarrow$ mean, std of $S_f[B]$
3. $Z_i \leftarrow 0.5\big(S_{s,i} - \mu_s\big)/(\sigma_s + \varepsilon) + 0.5\big(S_{f,i} - \mu_f\big)/\big(\sigma_f + \varepsilon\big)$
4. Sweep threshold τ on Z:
    - $\text{APCER}(\tau) = \text{mean}(Z_i < \tau \mid y_i = 1)$
    - $\text{BPCER}(\tau) = \text{mean}(Z_i \geq \tau \mid y_i = 0)$
5. EER ← APCER(τ^*) where |APCER − BPCER| is minimized.
6. BPCER@APCER=1% ← BPCER at the largest τ with APCER ≤ 0.01.
7. BPCER@APCER=5% ← Computed analogously.
8. ROC-AUC ← area under TPR vs FPR of Z.

Algorithm 4: Phase 1 Spatial Baseline (For Ablation)
This algorithm is identical to Algorithms 1–3, but with Stream B, the gate, the frequency head, and LoRA removed. It utilizes a single ArcFace head ($s = 30$, $m = 0.35$) on $e_{spatial}$ and is trained over four SMDD epochs.

5.0 Performance Analysis:

The evaluation of the UID-MAD system was conducted in two phases: a spatial-only baseline (Phase 1) and the proposed dual-stream calibrated system (Phase 2). All models were trained exclusively on the synthetic SMDD dataset and evaluated zero-shot on the FRLL-Morphs dataset across 15 different slices (five generators × three conditions).

Phase 1: Spatial-Only Baseline Results In-Domain Training (SMDD Dataset)
The Phase 1 spatial-only model converged rapidly on the synthetic training data, achieving near-perfect metrics by the fourth epoch.

Epoch	Stage	Train Loss	Train Acc
1	Head warmup	0.5422	92.06%
2	Head warmup	0.1544	95.54%
3	Head warmup	0.1226	96.17%
4	Fine-tune last 2 blocks	0.1223	96.15%

Zero-Shot Evaluation (FRLL Dataset - 15 Slices)
When tested on unseen real-world morphs, the Phase 1 spatial model successfully detected landmark-based morphs (OpenCV, FaceMorpher) but completely failed on generative latent morphs like StyleGAN2, performing no better than random chance (AUC ~0.50).

Generator	Condition	EER	BPCER@1%	BPCER@5%	ROC-AUC
OpenCV	Clean	**0.94%**	**0.0%**	0.0%	**0.9993**
OpenCV	JPEG (image)	**0.94%**	0.98%	0.0%	**0.9992**
OpenCV	Print-scan	9.16%	29.9%	15.7%	0.9786
FaceMorpher	Clean	**2.45%**	19.1%	0.0%	**0.9958**
FaceMorpher	JPEG (image)	2.49%	16.2%	0.0%	0.9956
FaceMorpher	Print-scan	5.85%	26.5%	7.4%	0.9859
WebMorph	Clean	4.91%	27.5%	4.9%	0.9886
WebMorph	JPEG (image)	5.85%	30.4%	7.4%	0.9875
WebMorph	Print-scan	25.4%	78.9%	58.3%	0.8320
AMSL	Clean	13.2%	76.0%	33.8%	0.9461
AMSL	JPEG (image)	13.7%	76.5%	32.4%	0.9434
AMSL	Print-scan	39.8%	92.2%	80.4%	0.6491
StyleGAN2	Clean	**51.0%**	**99.5%**	99.0%	**0.4988**
StyleGAN2	JPEG (image)	50.0%	99.5%	98.5%	0.4973
StyleGAN2	Print-scan	45.8%	96.1%	88.2%	0.5780

Phase 2: Dual-Stream Proposed Model (UID-MAD)
In-Domain Training (SMDD Dataset)
The dual-stream model, incorporating both spatial and frequency residual features, reached 0.00% validation EER and perfect ROC-AUC on the synthetic data by the second epoch of the head warmup stage.

Epoch	Stage	Train Loss	Train Acc	Val EER	Val ROC-AUC	Val Loss
1	Head warmup	2.7199	92.58%	0.0201%	0.999999	3.28×10^{-3}
2	Head warmup	1.4864	95.96%	**0.0000%**	**1.000000**	7.11×10^{-5}
3	Head warmup	1.1097	96.63%	0.0201%	0.999999	4.47×10^{-3}
4	Head warmup	0.8437	96.71%	**0.0000%**	**1.090000**	3.04×10^{-6}
5	LoRA fine-tune	0.6428	97.74%	**0.0000%**	**1.000000**	7.50×10^{-8}
6	LoRA fine-tune	0.5726	98.39%	**0.0000%**	**1.000000**	4.19×10^{-8}
7	LoRA fine-tune	0.5241	98.83%	**0.0000%**	**1.000000**	2.92×10^{-8}
8	LoRA fine-tune	**0.5039**	**99.12%**	**0.0000%**	**1.000000**	2.50×10^{-8}

Official Calibrated Results (FRLL Dataset - 15 Slices)
Using Z-score calibrated late fusion, the dual-stream model successfully recovered performance across all domains. The system achieved a mean calibrated ROC-AUC of approximately 0.992 across all 15 slices.

Generator	Condition	Calib. AUC	Calib. EER	BPCER@1%	Spatial AUC	Freq AUC	Fused-head AUC
OpenCV	Clean	**0.9997**	**0.74%**	**0.00%**	0.9991	0.8564	0.2937
OpenCV	JPEG	**0.9996**	**0.25%**	**0.00%**	0.9990	0.8576	0.2982
OpenCV	Print-scan	**0.9973**	**2.33%**	2.45%	0.9143	**0.9992**	0.6477
FaceMorpher	Clean	**0.9999**	**0.41%**	**0.00%**	0.9994	0.9506	0.2823
FaceMorpher	JPEG	**0.9999**	**0.49%**	**0.00%**	0.9993	0.9507	0.2838
FaceMorpher	Print-scan	**0.9994**	**1.02%**	1.47%	0.9342	**0.9996**	0.6580
WebMorph	Clean	**0.9968**	**1.96%**	14.71%	0.9963	0.7395	0.0324
WebMorph	JPEG	**0.9967**	**2.21%**	13.73%	0.9961	0.7426	0.0358
WebMorph	Print-scan	**0.9893**	**3.39%**	7.84%	0.6355	**0.9976**	0.2070
StyleGAN2	Clean	**0.9399**	**14.66%**	55.88%	0.9116	0.7644	0.1305
StyleGAN2	JPEG	**0.9384**	**14.33%**	56.37%	0.9083	0.7640	0.1348
StyleGAN2	Print-scan	**0.9854**	**3.92%**	17.65%	0.3155	**0.9977**	0.3952
AMSL	Clean	**0.9938**	**2.92%**	16.18%	0.9982	0.6605	0.3748
AMSL	JPEG	**0.9935**	**3.23%**	16.18%	0.9979	0.6634	0.3899
AMSL	Print-scan	**0.9809**	**6.73%**	14.22%	0.3743	**0.9981**	0.2259

Key Discoveries:

- **StyleGAN2 Resolution:** The dual-stream approach increased StyleGAN2 clean AUC from 0.4988 to 0.9399.
- **Print-Scan Orthogonality:** Print-scan analog filtering destroyed spatial performance (e.g., StyleGAN2 spatial AUC dropped to 0.3155), but it highly amplified the frequency stream's accuracy (frequency AUC rose to 0.9977), validating the dual-stream fusion design.

Comparison and Ablation

Intra-Project Ablation Study

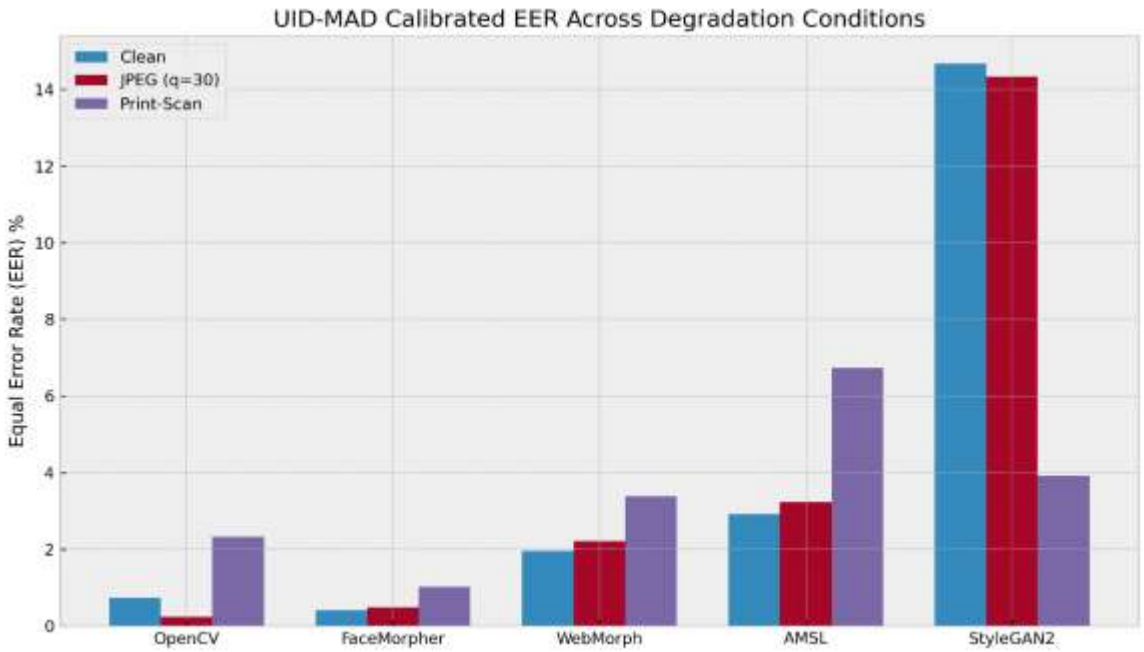
Comparing the different fusion strategies and streams confirms that the proposed Z-score calibrated fusion is the only variant that performs well across all conditions.

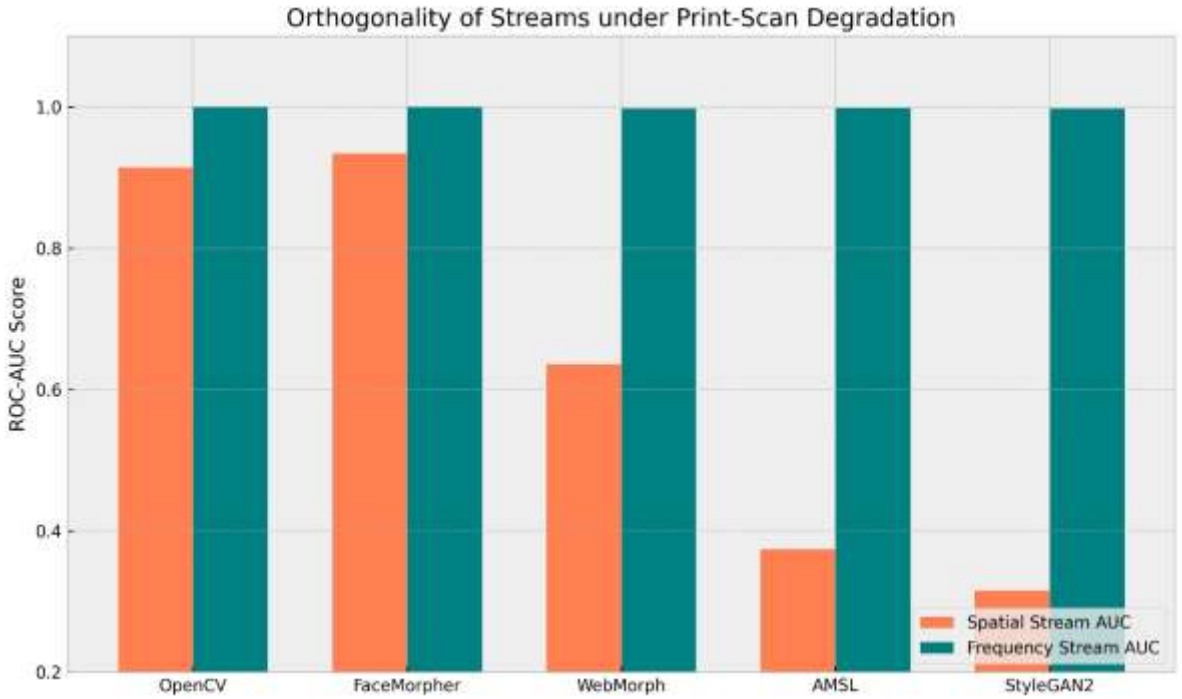
System	OpenCV Clean (AUC / EER)	StyleGAN2 Clean (AUC / EER)	StyleGAN2 Print-Scan AUC	Min AUC (over 15 slices)
Phase 1 spatial ViT + ArcFace	**0.9993 / 0.94%**	0.4988 / 51.0%	0.5780	0.4973
Phase 2 early concat	0.2422 (inverted)	**0.8525 / 24.0%**	—	collapsed landmark
Phase 2 max-OR softmax	0.7924 / 22.9%	0.5363 / 46.5%	0.6489	~0.51 (BPCER@1%=100%)
Phase 2 spatial head only	0.9991 / 0.53%	0.9116 / 17.0%	0.3155	0.3155
Phase 2 frequency head only	0.8564 / 23.0%	0.7644 / 29.9%	**0.9977**	0.6605 (AMSL clean)
Phase 2 fused ArcFace head	0.2937	0.1305	0.3952	0.0324
Phase 2 Z-score calibrated	**0.9997 / 0.74%**	**0.9399 / 14.66%**	**0.9854**	**0.9809**

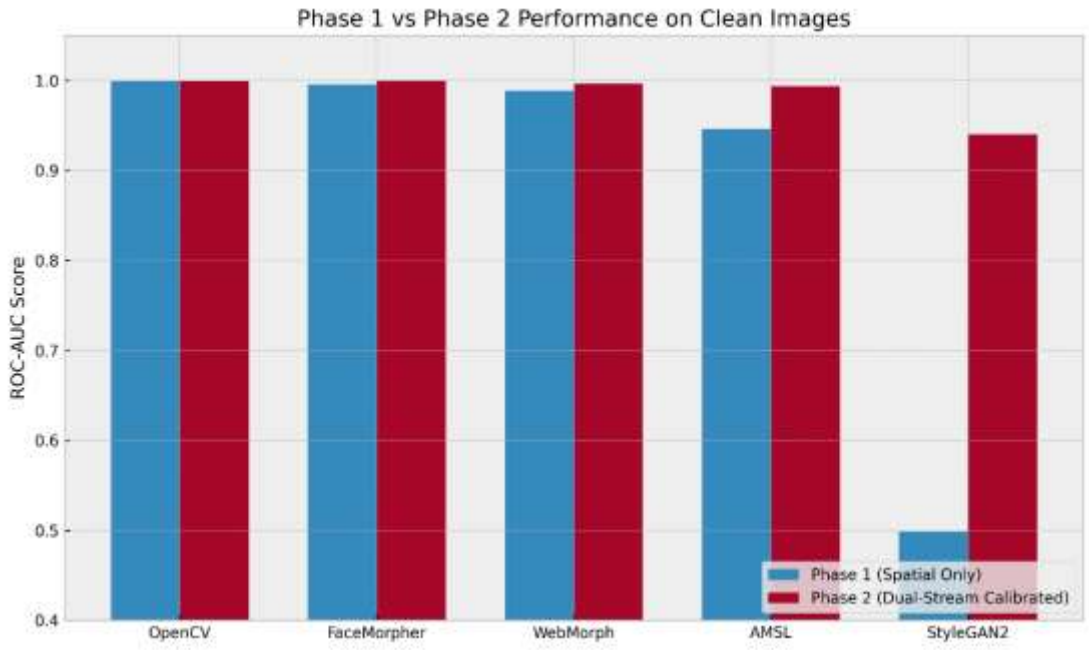
Phase 1 vs Phase 2 Performance Gains (Clean Images)

The addition of the frequency residual stream resulted in massive absolute gains for generative morphs while slightly improving already strong landmark morph detection.

Generator	Phase 1 Spatial AUC	Phase 2 Calibrated AUC	Absolute Gain
OpenCV	0.9993	0.9997	+0.0004
FaceMorpher	0.9958	0.9999	+0.0041
WebMorph	0.9886	0.9968	+0.0082
AMSL	0.9461	0.9938	+0.0477
StyleGAN2	0.4988	0.9399	**+0.4411**


UID-MAD Calibrated EER Across Degradation Conditions
Clean
JPEG (q=30)
Print-Scan
Equal Error Rate (EER) %
0
2
4
6
8
10
12
14
OpenCV
FaceMorpher
WebMorph
AMSL
StyleGAN2


Orthogonality of Streams under Print-Scan Degradation
ROC-AUC Score
0.2
0.4
0.6
0.8
1.0
OpenCV
FaceMorpher
WebMorph
AMSL
StyleGAN2
Spatial Stream AUC
Frequency Stream AUC


Phase 1 vs Phase 2 Performance on Clean Images
ROC-AUC Score
1.0
0.9
0.8
0.7
0.6
0.5
0.4
OpenCV
FaceMorpher
WebMorph
AMSL
StyleGAN2
Phase 1 (Spatial Only)
Phase 2 (Dual-Stream Calibrated)